

THREATLENS – PROJECT DETAILS & TECHNICAL SPECIFICATION

A Multi-Platform AI Cybersecurity & Real-Time Threat Intelligence Ecosystem

1. Project Domain

Cybersecurity / Mobile & Web Security / Artificial Intelligence & Machine Learning

ThreatLens is an enterprise-grade, multi-platform cybersecurity intelligence ecosystem engineered to protect modern users against sophisticated digital threats across both mobile (Android) and desktop (Chromium) platforms. The project integrates computer vision, on-device deterministic heuristics, multi-engine threat intelligence feeds, and cloud-assisted Large Language Model (LLM) reasoning to detect and neutralize malicious QR codes (Quishing), zero-day phishing URLs, malware distribution binaries, intrusive advertising telemetry, and brand impersonation attacks in real time.

2. Problem Statement

A QR code itself is not harmful; the danger lies in the destination website it forces your device to open.

Most standard smartphone cameras and basic QR scanner apps suffer from a major security flaw: they open scanned links automatically without checking where the link actually leads. Cybercriminals actively exploit this through a technique known as **Quishing (QR Phishing)**. Attackers print fraudulent QR stickers and paste them over legitimate QR codes at parking meters, restaurant tables, payment counters, transit stations, and in scam letters. When an unsuspecting user scans the code, their phone immediately opens a website designed to steal account passwords, trigger unauthorized payment transfers, or silently download malicious apps and viruses.

Furthermore, everyday web browsing on desktop computers faces similar risks through shortened links, hidden redirect chains, drive-by executable downloads (.exe, .scr, .apk), and invasive tracker networks. Most existing security tools only show an opaque 'Safe' or 'Dangerous' label without explaining *why* in plain language, leaving users uninformed.

Therefore, there is an urgent need for an all-in-one, user-friendly security system that acts as an active firewall for QR codes and web links—inspecting, explaining, and neutralizing threats before any harm can occur.

3. Project Objectives

Objective 1: Real-Time Mobile QR Security Scanning (Android)

To build a fast and simple Android app using camera scanning (Google ML Kit) that reads QR codes instantly and checks their safety in under 50 milliseconds before opening them.

Objective 2: Pre-Navigation Link Inspection & Redirect Unrolling

To automatically expand shortened URLs (such as bit.ly), trace multi-step redirect chains, and apply smart on-device heuristic rules to catch fake lookalike domains (e.g., g00gle), direct IP addresses, and suspicious web extensions.

Objective 3: Clear and Explainable AI Threat Analysis

To combine real-time threat intelligence databases (Google Safe Browsing, VirusTotal, AbuseIPDB) with artificial intelligence (LLM7.io Cloud AI) to provide an easy-to-understand Trust Score (0 to 100) and plain-English explanations of detected risks.

Objective 4: Safe Isolated Browsing & Child Safety Controls

To provide a built-in isolated Sandbox Browser that blocks intrusive ads, trackers, and harmful scripts, while offering simple parental controls to protect children from adult content, bedtime screen overuse, and online scams.

Objective 5: Desktop Web Shielding & Certified Safe QR Codes

To protect desktop users with a browser extension that guards against suspicious link clicks and dangerous downloads, while allowing users to create tamper-proof 'Certified Safe QR Codes' that recipients can verify and trust.

4. Dataset Name and Source

Dataset Name: Phishing, Malware & Threat Intelligence Benchmark Dataset

Active Threat Intelligence Feeds & Data Sources:

- **PhishTank:** Validated real-time phishing URL feeds, target brand labels, and community submission timestamps.
- **OpenPhish:** Zero-day phishing intelligence feeds and algorithmic threat indicators.
- **URLhaus (abuse.ch):** Active malware distribution URLs, malicious payload hashes (.exe, .apk, .bat, .dll), and threat tags.
- **Google Safe Browsing v4:** Real-time global lists of suspected phishing, malware, and unwanted software destinations.

- **VirusTotal Multi-Engine Intelligence:** Consolidated reputation metrics aggregated from 70+ antivirus and domain security engines.
- **AbuseIPDB & WhoisXML:** IP abuse confidence scores, ASN reputation, domain registration age, and SSL certificate validity.
- **EasyList & EasyPrivacy Rulesets:** Comprehensive declarative rulesets for tracking and intrusive advertising domains.
- **Synthetic Quishing Benchmark:** Curated dataset of obfuscated QR matrices, homoglyph domains, and authority spoofing patterns.

Note: The datasets and feeds directly power the deterministic heuristic rules and model weights in the active implementation.

5. Algorithms and Detection Architecture

ThreatLens implements a hybrid multi-tiered detection pipeline designed for sub-second response times without sacrificing accuracy:

1. **Deterministic Lexical & Structural Heuristic Engine:** Inspects URL entropy, domain length, subdomain depth (flagging >3 parts), Cyrillic/homoglyph lookalikes (Punycode *xn--*), direct IP addressing, '@' authority section credential harvesting, and suspicious TLDs (.xyz, .top, .pw, .buzz, .gripe, .bet).
2. **Machine Learning URL & Content Classifiers:** Random Forest and Gradient Boosted Decision Trees trained on lexical, DNS, and structural features for high-precision binary classification (Safe vs. Malicious). Naive Bayes and Logistic Regression provide fast triage for SMS/text smishing detection.
3. **TF-IDF & Character N-Gram Feature Extraction:** NLP feature extraction capturing obfuscated brand names (e.g., *paypa1*, *g00gle*, *sec-chase*) and deceptive text phrases within landing pages.
4. **Large Language Model (LLM) Reasoning Engine:** Utilizes the **LLM7.io Cloud AI API** to synthesize human-readable security insights, evaluate attacker intent, and explain risk scores transparently.
5. **Computer Vision & Matrix Decoding:** **Google ML Kit Barcode Scanning API** (Android) and **jsQR OffscreenCanvas** (Extension) for instant detection and rectification of distorted, low-contrast, or inverted QR matrices.
6. **Declarative Net Request (DNR) Sub-Millisecond Filtering:** High-performance network-level rule matching against 30,000+ compiled rules neutralizing ads, telemetry trackers, and malware hosts before socket connection establishment.

6. Tools and Technologies

Mobile Platform (Android):

- **Language:** Kotlin 2.0+ (100% Kotlin codebase)
- **UI Framework:** Jetpack Compose, Material Design 3, Dynamic Light/Dark Theme
- **Concurrency & Architecture:** Kotlin Coroutines, StateFlow, ViewModel, Clean MVVM
- **Vision & Hardware:** CameraX, Google ML Kit Barcode Scanning API
- **Web Sandbox:** Android WebView, Custom WebViewClient with HTTP 204 Ad Interception
- **Build & Tooling:** Gradle 8.13, Kotlin Symbol Processing (KSP)

Browser Extension (Desktop):

- **Platform:** Chromium Manifest V3 (MV3)
- **Languages:** JavaScript (ES6+), HTML5, Vanilla CSS3 (Loanzo Light Cyber HUD)
- **Core APIs:** Declarative Net Request (DNR), OffscreenCanvas, Downloads API, Tabs, ContextMenus, Storage Local
- **Key Engines:** HeuristicsEngine.js, UrlExpander.js, dom-scanner.js, surrogates.js, jsQR.js

AI, ML & Backend Infrastructure:

- **Languages:** Python 3.10+, Node.js (v18+)
- **Libraries & Frameworks:** PyTorch, Scikit-learn, Pandas, NumPy, ONNX Runtime, FastEmbed
- **LLM Inference:** LLM7.io Cloud AI API / Neural Threat Reasoning Engine
- **Cloud Services:** Firebase Authentication, Cloud Firestore (cross-device sync), Firebase Cloud Functions, Vercel Serverless

External Security APIs:

- Google Safe Browsing v4, VirusTotal v3, URLScan.io, AbuseIPDB, Cloudflare Radar, WhoisXML

Development and Version Control:

- Git, GitHub, Android Studio Ladybug / Meerkat, VS Code, Google Antigravity IDE

7. Expected Outcome and Deliverables

The completed ThreatLens project delivers a fully operational, multi-platform cybersecurity system:

- **Production Android Application (app-debug.apk):** Features real-time ML Kit QR scanning, comprehensive threat report cards with 0–100 Trust Scoring, neural AI insights, an isolated Sandbox Browser with ad/tracker blocking, parental controls, and certified QR generation.
- **Enterprise Manifest V3 Chrome Extension:** Features pre-navigation Link Guard with animated cyber radar rings, multi-ruleset Declarative Net Request ad/malware shielding, screen-snip QR decoding (Alt+Q), executable download interception, and uBlock/Brave scriptlet surrogates.
- **Cross-Device Cloud Synchronization:** Real-time sync of scan history, custom whitelist domains, and community threat reports via Cloud Firestore.
- **Public Showcase & Live Analyzer Portal:** Production web portal with responsive Loanzo light design, interactive URL scanner demonstration, and distribution links.
- **User Awareness & Threat Literacy:** Elevates user cybersecurity awareness through transparent, explainable threat breakdowns rather than black-box verdicts.

The system acts as an active threat-interception and awareness platform, providing defense-in-depth across mobile and web attack vectors.

8. Relevant Research Papers

Research Paper 1

Title: Machine Learning Based Phishing Detection from URLs

Authors: Sahingo, O. K., Buber, E., Diri, B.

Publication: Expert Systems with Applications, 2019

Relevance: Provides empirical foundations for real-time lexical and structural feature extraction in high-throughput URL classification.

DOI / Reference: <https://doi.org/10.1016/j.eswa.2018.09.029>

Research Paper 2

Title: PhishNot: A Cloud-Based Machine-Learning Approach to Phishing URL Detection

Authors: Alani, M. M., & Tawfik, H.

Publication: Computer Networks, 2022

Relevance: Evaluates cloud-assisted Random Forest classifiers and lightweight edge inference models for phishing URL detection.

DOI / Reference: <https://doi.org/10.1016/j.comnet.2022.109153>

Research Paper 3

Title: QR Code Security and Vulnerabilities (Quishing): A Systematic Survey

Authors: Vidas, T., et al.

Publication: IEEE Security & Privacy, 2021

Relevance: Analyzes visual barcode spoofing, URL obfuscation in physical vectors, and user deception models in mobile QR scanning.

DOI / Reference: <https://doi.org/10.1109/MSP.2021.3082914>

Research Paper 4

Title: Applications of Deep Learning for Phishing Detection: A Systematic Literature Review

Authors: Catal, C., Giray, G., Tekinerdogan, B.

Publication: Knowledge and Information Systems, 2022

Relevance: Comprehensive review of DNN, CNN, and transformer architectures for detecting zero-day social engineering and malicious URLs.

DOI / Reference: <https://doi.org/10.1007/s10115-022-01683-1>

Research Paper 5

Title: Phishing URL Detection with Neural Networks: An Empirical Study

Authors: Benavides, E., et al.

Publication: Scientific Reports, 2024

Relevance: Empirical benchmarking of neural network feature extraction using PhishTank and OpenPhish live threat datasets.

DOI / Reference: <https://doi.org/10.1038/s41598-024-54123-x>

Research Paper 6

Title: Securing the Modern Web: Declarative Content Filtering in Manifest V3

Authors: Reis, C., et al.

Publication: ACM Conference on Computer and Communications Security (CCS), 2023

Relevance: Explores privacy-preserving Declarative Net Request rulesets, scriptlet surrogates, and sub-millisecond network interception in modern browsers.

DOI / Reference: <https://doi.org/10.1145/3576915.3623101>